CONE TRIANGLES OF ADMISSIBLE SHORT EXACT SEQUENCES ARE HOMOTOPY EQUIVALENT TO THE QUOTIENT, AND ARBITRARY MORPHISMS REDUCE TO ADMISSIBLE MONOMORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the comparison map C(alpha) to M with its homotopy inverse (s, -delta), the differential compatibility via the characterizing property of delta, the displayed homotopy identity with the explicit homotopy h, the cone-map compatibility check p after (s, -delta) = -delta, and the admissible-replacement reduction of part (2) through lemma-make-injective, lemma-functorial-cone and lemma-third-isomorphism are all retained verbatim.

Editorial note. The label lemma-the-same-up-to-isomorphisms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Termwise split sequences and the characterizing differential. Homology, Lemma homology-lemma-ses-termwise-split-cochain. Supplies the characterization of delta as the unique map with d after s minus s after d equal to alpha after delta, which makes the candidate homotopy inverse (s, -delta) compatible with differentials.
- Admissible replacement of a morphism by a monomorphism. Same chapter, Lemmas lemma-make-injective, lemma-functorial-cone and lemma-third-isomorphism. Reduces part (2) to part (1): after replacing L by the admissible supertilde L, the two cone triangles are isomorphic, so f may be assumed an admissible monomorphism.

Source reference. Original reference 014E: sources/stacks-dga/pinned-original-source.tex, source lines 619–619. Prerequisite pointer invoked at proof line(s) 1189; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 011J: sources/stacks-chow/tags.txt, source lines 1365–1365. Prerequisite pointer invoked at proof line(s) 1115; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 014F: sources/stacks-dga/pinned-original-source.tex, source lines 644–644. Prerequisite pointer invoked at proof line(s) 1202; it names a

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 013N: sources/stacks-dga/pinned-original-source.tex, source lines 777–777. Prerequisite pointer invoked at proof line(s) 1201; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 014K: sources/stacks-dga/pinned-original-source.tex, source lines 1016–1016. Prerequisite pointer invoked at proof line(s) 1202; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let (A, d) be a differential graded algebra.*

(1) *Given an admissible short exact sequence $0 \to K \xrightarrow{\alpha} L \to M \to 0$ of differential graded A-modules there exists a homotopy equivalence $C(\alpha) \to M$ such that the diagram*

$$\begin{array}{ccccccc} K & \longrightarrow & L & \longrightarrow & C(\alpha) & \xrightarrow[-p]{} & K[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ K & \xrightarrow{\alpha} & L & \xrightarrow{\beta} & M & \xrightarrow{\delta} & K[1] \end{array}$$

defines an isomorphism of triangles in $K(Mod_{(A,d)})$.

(2) *Given a morphism of complexes $f : K \to L$ there exists an isomorphism of triangles*

$$\begin{array}{ccccccc} K & \longrightarrow & \tilde{L} & \longrightarrow & M & \xrightarrow[\delta]{} & K[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ K & \longrightarrow & L & \longrightarrow & C(f) & \xrightarrow{-p} & K[1] \end{array}$$

where the upper triangle is the triangle associated to a admissible short exact sequence $K \to \tilde{L} \to M$.

Proof. Proof of (1). We have $C(\alpha) = L \oplus K$ and we simply define $C(\alpha) \to M$ via the projection onto L followed by β. This defines a morphism of differential graded modules because the compositions $K^{n+1} \to L^{n+1} \to M^{n+1}$ are zero. Choose splittings $s : M \to L$ and $\pi : L \to K$ with $\mathrm{Ker}(\pi) = \mathrm{Im}(s)$ and set $\delta = \pi \circ \mathrm{d}_L \circ s$ as usual. To get a homotopy inverse we take $M \to C(\alpha)$ given by $(s, -\delta)$. This is compatible with differentials because δ^n can be characterized as the unique map $M^n \to K^{n+1}$ such that $\mathrm{d} \circ s^n - s^{n+1} \circ \mathrm{d} = \alpha \circ \delta^n$, see proof of Homology, Lemma 011J. The composition $M \to C(f) \to M$ is the identity. The composition $C(f) \to M \to C(f)$ is equal to the morphism

$$\begin{pmatrix} s \circ \beta & 0 \\ -\delta \circ \beta & 0 \end{pmatrix}$$

To see that this is homotopic to the identity map use the homotopy $h : C(\alpha) \to C(\alpha)$ given by the matrix

$$\begin{pmatrix} 0 & 0 \\ \pi & 0 \end{pmatrix} : C(\alpha) = L \oplus K \to L \oplus K = C(\alpha)$$

It is trivial to verify that

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} s \\ -\delta \end{pmatrix} \begin{pmatrix} \beta & 0 \end{pmatrix} = \begin{pmatrix} \mathrm{d} & \alpha \\ 0 & -\mathrm{d} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ \pi & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ \pi & 0 \end{pmatrix} \begin{pmatrix} \mathrm{d} & \alpha \\ 0 & -\mathrm{d} \end{pmatrix}$$

To finish the proof of (1) we have to show that the morphisms $-p : C(\alpha) \to K[1]$ (see Definition 014E) and $C(\alpha) \to M \to K[1]$ agree up to homotopy. This is clear from the above. Namely, we can use the homotopy inverse $(s, -\delta) : M \to C(\alpha)$ and check instead that the two maps $M \to K[1]$ agree. And note that $p \circ (s, -\delta) = -\delta$ as desired.

Proof of (2). We let $\tilde{f} : K \to \tilde{L}$, $s : L \to \tilde{L}$ and $\pi : L \to L$ be as in Lemma 013N. By Lemmas 014F and 014K the triangles $(K, L, C(f), i, p)$ and $(K, \tilde{L}, C(\tilde{f}), \tilde{i}, \tilde{p})$ are isomorphic. Note that we can compose isomorphisms of triangles. Thus we may replace L by $\tilde{L}$ and f by $\tilde{f}$. In other words we may assume that f is an admissible monomorphism. In this case the result follows from part (1). $\square$